

Reproductive and Gonadal-Endocrine Emergence on a Jammed-Vacuum Substrate: the HPG Oscillator, the Menstrual Cycle, and Hormone-Driven Cancer

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Abstract

We derive the human reproductive and gonadal-endocrine (hypothalamic–pituitary–gonadal, HPG) axis on the VP Theory jammed-vacuum substrate, using two vendored primitives unchanged: a bistable rectification (R19) switch and a FitzHugh–Nagumo (FHN) relaxation oscillator. A single sex-hormone drive h acting on the R19 tilted double well produces *two* opposite failure modes — an oscillator disease and a barrier-lowered oncogenic switch — governed by *one* control axis: temporal pattern (pulsatile versus continuous, cycling versus sustained). The same kernel, run as an FHN oscillator, generates the GnRH pulse train, the menstrual cycle, and the spermatogenic cycle by changing only the recovery timescale τ_s ; run as an R19 switch, it gives organ emergence in measured- γ order, the discontinuous ovulatory LH surge, puberty as a spinodal crossing, and hormone-driven carcinogenesis as barrier lowering. The same two primitives extend downstream to the gamete and the embryo, to the infertility/subfertility distinction (categorical past a spinodal versus a finite Kramers rate near threshold), and to sex determination and sex-ratio distortion (a tiltable bistable switch with Fisher’s 1:1 restoring force). Every published quantity is pulled live from a deterministic reproduction engine (SEED = 19; two builds are byte-identical, sha256 = f849da7d13d23302...); no constant is tuned to a target. Grades are honest throughout: [F] forced by geometry, [V] simulation-verified, [L] measured anchor, [O] open with the obstacle stated. This is a falsifiable physics-derived research artefact and is **not** clinical advice.

Keywords: reproductive endocrinology; HPG axis; GnRH pulse-frequency coding; menstrual cycle; relaxation oscillator; FitzHugh–Nagumo; bistable switch; spinodal; hysteresis; hormone-driven cancer; dose–response; Bipolar Androgen Therapy; infertility; subfertility; meiotic drive; sex determination; sex ratio; Fisher’s principle; VP Theory; jamming; computational biology; reproducibility.

1 Scope and the jammed-vacuum substrate

One drive, two failures, one lever. This package derives the gonadal-endocrine axis on the VP Theory jammed-vacuum substrate [1], where a cell or tissue fate is a point in a bistable R19 double-well potential and a hormone is a tilt h on that well. One sex-hormone drive, moved two opposite ways, produces the whole disease map: move it the wrong *pattern* and an HPG oscillator fails (an endocrine disorder); lower the barrier and the R19 oncogenic switch crosses (a hormone-driven cancer). The single control axis the framework exposes is therefore temporal pattern. No new physics is introduced here; the vendored primitives are used unchanged.

The fate landscape is the tilted double well

$$V(s) = -\frac{1}{2}\gamma s^2 + \frac{1}{4}s^4 - h s, \quad \dot{s} = \gamma s - s^3 + h, \quad (1)$$

whose stationary states are the roots of $s^3 - \gamma s - h = 0$: for three real roots there are two minima (basins) and a saddle. As the tilt h grows, the metastable minimum and the saddle

approach and annihilate at the *spinodal* — beyond it the flip is unavoidable and discontinuous. Two locked constants set every threshold and barrier in the paper, both forced functions of the identity constant γ alone:

$$h_{\text{sp}}(\gamma) = 2 \left(\frac{\gamma}{3}\right)^{3/2} \quad (= 0.3849 \text{ at } \gamma = 1), \quad \Delta V_0(\gamma) = \frac{\gamma^2}{4}. \quad (2)$$

The spinodal h_{sp} fixes where a switch flips; the barrier ΔV_0 fixes how stable a basin is. Both are **[F]** forced by geometry.

Firewall. Decomposition is by physical regime, not textbook organ labels: this is the jamming (hormonal-cycle / germline) class. The brain-facing hypothalamic–pituitary–adrenal (HPA) stress axis and felt experience belong to the companion Felt Cognition paper and are cited, never re-derived here. Sex-hormone *levels* are this package’s single source of truth.

2 Organ emergence from measured master-gene γ

The morphogenesis identity constant is $\gamma = -\langle \Delta G_{37}^{\text{NN}} \rangle$: minus the mean nearest-neighbour stacking free energy [3] over the proximal promoter (TSS–2000... + 500, GRCh38) of each organ’s master gene. It is **[V]** measured, never fitted — the pipeline is accepted only because it reproduces the vendored SOX9 anchor ($\gamma = 1.4598$) to four decimals; the promoter sequences are cached so γ reproduces offline bit-for-bit [2]. The emergence order of an organ system is the argsort of its master-gene γ values, the same gene-clock used elsewhere in the programme.

For the four reproductive masters the measured order (ascending γ) is germline $\rightarrow$ testis $\rightarrow$ ovary $\rightarrow$ tract. Germline-first is consistent with primordial germ-cell specification preceding gonadal differentiation, a sign sanity-check rather than a fitted result. The relative run-length / size of each switch follows DWELL(γ) = $\gamma^{3/2}/(K + \text{brake})$ — order and direction **[F]** forced; absolute magnitude **[O]** (external calibration; see §16).

Table 1: Reproductive organ emergence in measured- γ order. γ is measured from the proximal promoter by the 4D DNA pipeline, reproduced offline bit-for-bit; the emergence order is argsort(γ).

master gene	organ	γ (meas.)	func. spinodal	rel. size (DWELL)	dyn. class
DAZL	germline	1.3803	0.6242	1.4742	germline
SOX9	gonad testis	1.4598	0.6789	1.6034	germline-support
FOXL2	gonad ovary	1.4829	0.6950	1.6416	oscillator
WT1	reproductive tract	1.5182	0.7200	1.7006	structural

3 The GnRH pulse generator

Answer. The hypothalamic GnRH pulse generator is the shared FHN relaxation oscillator: a fast R19 switch with slow recovery emits a relaxation pulse train (62 pulses at $\tau_s = 60$). Pulse *frequency* — not amplitude — decodes to gonadotropin identity, fast pulses favouring LH and slow pulses favouring FSH, reproduced as a monotone function of τ_s [7]. The mechanism is **[V]** sim-verified; the absolute pulse rate is a measured **[L]** anchor.

The arcuate KNDy network that paces GnRH release is, on this substrate, the same fast–slow FHN system used everywhere in the package. At the follicular setting $\tau_s = 60$ the generator emits 62 pulses over the run with mean inter-pulse interval 130.6 (arb). The clinically central fact — that pulse *frequency* sets the LH:FSH balance — falls out of the recovery timescale: a faster generator (smaller τ_s) fires more often and favours LH; a slower generator favours FSH;

and the pulse rate is monotone in τ_s (Table 2). The pulse rate is calibrated to the measured follicular figure of roughly one GnRH/LH pulse per 60–90 min [L]; the package does not claim to derive that clinical number, only the monotone frequency-decoding mechanism [V].

Table 2: Frequency decoding by the shared FHN generator: the pulse rate is a monotone function of the recovery timescale τ_s , and fast/slow maps onto the LH/FSH bias.

recovery τ_s	pulses	mean interval	pulse rate	gonadotropin bias
20	155	51.7	0.019375	LH (fast)
60	62	130.6	0.007750	LH (fast)
150	27	299.4	0.003375	FSH (slow)
400	11	756.2	0.001375	FSH (slow)

4 The menstrual cycle as a relaxation oscillator

Answer. The menstrual cycle is the same kernel run slow: a relaxation (saw-tooth) oscillator, not a sinusoid. Over the run it completes 11 cycles with period 1133.7 (arb), built from a slow follicular rise (3.6) and an abrupt luteal fall (1130.0), an asymmetry ratio of 309.6. The strong rise/fall asymmetry is the signature of a relaxation oscillation; a sinusoid would have asymmetry near unity. The period magnitude is anchored to the ~ 28 -day cycle as a measured [L] quantity, while the relaxation *form* is [V] sim-verified. Only the recovery timescale τ_s distinguishes this from the GnRH pulse and the spermatogenic cycle — it is one oscillator at three speeds.

5 The oestrogen feedback switch and the LH surge

Answer. The pre-ovulatory LH surge is a discontinuous spinodal crossing of the R19 switch, not a smooth rise: rising oestradiol tilts the well until the negative-feedback basin annihilates at $h_{sp} = 0.3849$, and the state jumps to the secretory branch. The model shows a discontinuous up-jump of size 1.634 once the up-threshold 0.3873 is crossed, and the return path crosses a different down-threshold -0.3915 — a hysteresis loop of width 0.779 [V]. This is the switch from oestrogen *negative* feedback (most of the cycle) to the brief *positive*-feedback surge: the substrate realises it as one bistable well crossing its spinodal in one direction, then returning by another, which is exactly what hysteresis means. The discontinuity and the hysteresis are forced features of an R19 switch, not added by hand.

6 The spermatogenic cycle

Answer. Spermatogenesis is the same relaxation oscillator at an intermediate recovery timescale ($\tau_s = 340$), giving period 656.5 (arb), between the fast GnRH pulse and the slow menstrual cycle. The package verifies the period ordering directly: 132.4 (pulse) < 656.5 (spermatogenic) < 1133.7 (menstrual) [V]. Mapped to clinical time the spermatogenic cycle anchors to ~ 16 days [L]; the ordering is the derived claim, the absolute days an anchor. That a single oscillator covers three reproductive rhythms purely by its recovery timescale is the unifying point: the body re-uses one fast-slow kernel and re-tunes τ_s .

7 Puberty as a spinodal crossing

Answer. Puberty is the reactivation of the GnRH generator modelled as a single irreversible spinodal crossing: below threshold the axis sits in the quiescent (OFF) basin at -0.514 , and

once the drive reaches 0.3888 — a ratio jump/spinodal of 1.01 — the state jumps by 0.955 to the active (ON) branch at 0.442 [V]. The onset is therefore discontinuous and localised at the spinodal, the same geometry as the LH surge but taken once and not returned. The framework fixes the discontinuous order (OFF below the crossing, ON above) and the jump/spinodal ratio; mapping the crossing to a chronological age requires external endocrine calibration and is declared [O] (§16).

8 Hormone-driven cancer as barrier lowering

Answer. Hormone-driven carcinogenesis is the R19 oncogenic switch crossed by barrier lowering: a sustained sex-hormone drive tilts the well and lowers the inter-basin barrier $\Delta V_0 = \gamma^2/4$, raising the Kramers escape rate [6] into the malignant basin. The relative risk is the ratio of escape rates,

$$\text{RR}(\text{dose}) = \frac{\exp[-\Delta V(h)/D]}{\exp[-\Delta V(0)/D]}, \quad (3)$$

with a single open lattice noise constant $D = 0.15$ that sets the absolute magnitude only. Crucially, the relative barrier as a function of relative drive is *identical across* γ (Table 3, max spread 0.0), so the dose–response *shape* is forced and γ -independent [F]; every reported RR *ratio* is invariant under D (verified D -robust). Three cancers come out as three distinct, sim-verified shapes:

- **Breast** (Table 4): cumulative-oestrogen exposure raises RR monotonically to a plateau (saturating once the barrier is gone) [V].
- **Prostate** (Table 5): barrier reduction is concave and RR plateaus past the fold [V].
- **Cervical** (Table 6): HPV $\times$ smoking is multiplicative at low exposure and saturates sub-multiplicatively at high exposure [V], the same synergy structure seen in other infection $\times$ carcinogen pairs.

Absolute population incidence needs epidemiological calibration and is declared [O]; the shapes, ratios, and the γ -independence are the derived content.

Table 3: Barrier-law universality: the relative barrier as a function of relative drive is identical across γ (max spread 0.0), so the dose–response shape is γ -independent [F].

h/h_{sp}	$\Delta V/\Delta V_0$	max spread across γ
0.00	1.00000	0.0
0.20	0.71014	0.0
0.40	0.45804	0.0
0.60	0.24771	0.0
0.80	0.08705	0.0
0.95	0.01083	0.0

Table 4: Breast: cumulative-oestrogen exposure monotonically raises modelled relative risk to a plateau. Population anchor: WHI combined MHT RR ~ 1.26 [L].

exposure yr	h/h_{sp}	RR
0	0.000	1.000
5	0.312	2.067
10	0.624	3.633
15	0.935	5.156
20	0.999	5.294
25	0.999	5.294
30	0.999	5.294

Table 5: Prostate: barrier reduction is concave and the relative risk plateaus past the fold.

androgen drive $\times h_{\text{sp}}$	barrier reduction	RR
0.0	0.000	1.000
0.2	0.290	1.621
0.4	0.542	2.468
0.6	0.752	3.504
0.8	0.913	4.580
1.0	1.000	5.295
1.3	1.000	5.295
1.6	1.000	5.295

Table 6: Cervical: HPV $\times$ smoking is multiplicative at low exposure (synergy index ≈ 1) and saturates sub-multiplicatively at high exposure.

joint exp.	RR HPV	RR smoke	RR both	synergy idx
0.05	1.224	1.135	1.376	0.990
0.10	1.473	1.283	1.819	0.963
0.20	2.030	1.621	2.853	0.867
0.30	2.636	2.017	3.929	0.739
0.40	3.248	2.468	4.825	0.602

9 Temporal pattern as a therapeutic lever

Answer. The single control axis the framework exposes is temporal pattern, and it runs the whole therapy map. Cycling the androgen drive (Bipolar Androgen Therapy, BAT) amplifies the stress on the resistant clone $13.6\times$ over continuous high suppression (Table 7): a resistant clone adapted to a constant environment is destabilised by forced oscillation, whereas continuous suppression selects for it. The same principle, read on the GnRH generator, gives the textbook paradox that the *same molecule* has opposite effects by pattern: pulsatile GnRH activates the axis (41 pulses) while continuous GnRH suppresses it (0 pulses) [V]. This retrodicts pulsatile-GnRH pump therapy (activation in hypothalamic amenorrhoea) and depot GnRH-agonist desensitisation (suppression in precocious puberty, endometriosis, and prostate cancer), the hCG trigger as a single spinodal kick onto the ON branch, and the BAT schedule — all as moves along one temporal-pattern axis.

Table 7: Temporal-pattern selectivity. Cycling androgen (Bipolar Androgen Therapy) amplifies the stress on the resistant clone far more than continuous suppression does.

schedule	sensitive stress	resistant stress	res./sens.
continuous high (T)	0.0036	0.0304	8.35
continuous low (ADT)	0.0018	0.0043	2.36
cycling (BAT)	0.0633	0.4125	6.52

10 Disease mechanisms on one substrate

Answer. Eight common HPG disorders map to one substrate: one drive, two failure modes, one lever. PCOS (the generator stuck too fast, LH-biased rate 0.0135) and functional hypothalamic amenorrhoea (the same generator slowed below the ovulatory rate, 0.0025) are opposite ends of one pulse-frequency axis. Menopause is irreversible loss of the bistable secretory latch (ON $1.0 \rightarrow 0.229$ on follicular depletion), so HRT is replacement, not restoration. The split is by mechanism, not body part: every disorder is either a broken oscillator or a crossed oncogenic switch, resolved by the temporal-pattern lever.

Retrodictions are sim-verified [V]; the genuinely novel suggestions — low-frequency pulsatile restoration in PCOS, and pulsed/cycled testosterone replacement — are graded [O] and explicitly require clinical validation.

Table 8: Eight common HPG disorders on one substrate. Retrodictions of existing practice are flagged as validation; genuinely novel hypotheses are graded [O] and require clinical validation. None of this is clinical advice.

disorder	substrate failure mode	model-derived hypothesis	grade
PCOS (anovulatory phenotype)	GnRH pulse generator runs too FAST -> chronic LH excess, low FSH -> follicles recruited but none selected, no dominant follicle, no surge (chronic anovulation)	restore the SLOW pulse pattern itself – low-frequency PULSATILE GnRH (not continuous, not fast) to re-establish the FSH-favouring rhythm and physiological follicle selection	[O]
Functional hypothalamic amenorrhoea (FHA)	energy deficit / stress SLOW the GnRH pulse generator below the ovulatory pulse frequency (PCOS and FHA are opposite ends of one pulse-frequency axis); reversible	pulsatile GnRH at PHYSIOLOGICAL frequency restores the generator's rate – the model says the missing thing is the pulse PATTERN/rate, not the molecule	[V]
Menopause / perimenopause	follicle pool (ovarian responder substrate) exhausted -> the bistable secretory latch is lost; irreversible substrate depletion, NOT detuning	the model is explicit that this is REPLACEMENT, not restoration: the oscillator hardware is gone, so the honest in-framework lever is replacement with the cancer-risk trade-off made explicit (same h that lowers the breast barrier) – restoration would need follicle regeneration (out of scope)	[O]
Age-related male hypogonadism	sustained androgen drive decays with age -> hypogonadal symptoms	TRT re-supplies the SAME androgen drive that lowers the prostate R19 barrier (trade-off). The temporal-pattern lever predicts that PULSATILE or cycled delivery might separate the wanted androgenic tone from the unwanted sustained proliferative drive	[O]
Central precocious puberty	the GnRH axis crosses the T5 spinodal too EARLY -> premature pulsatile reactivation	retrodictio: continuous GnRH agonist = depolarisation block of the generator (same mechanism the therapy module shows) -> axis switched OFF; the standard of care is exactly the substrate-predicted move	[V]
Delayed puberty (functional / constitutional)	drive has not yet crossed the T5 spinodal -> axis still OFF	supply drive to cross the spinodal; pulsatile GnRH drives the generator (activation pattern) where continuous would block it	[V]
Anovulatory infertility (surge failure)	the mid-cycle LH-surge switch fails to flip -> no ovulation	retrodictio: the hCG trigger is a single SPINODAL KICK forcing the surge switch onto the ON branch – the model identifies why a one-shot trigger (not sustained drive) is the right move at that moment	[V]
Endometriosis (oestrogen-dependent)	oestrogen-driven EC-TOPIC proliferation – SAME drive variable as breast carcinoma, benign	suppression is the goal here, so the model says CONTINUOUS (block) is correct and warns that a cycling/pulsatile	[V]

11 The gamete itself: meiosis, motility and the egg

Answer. The four questions about the germ cell answer on the same substrate, with measured gamete-program promoter γ as the only new input. *How* a gamete is made: meiosis is one DNA replication then two divisions ($2N, 2C \rightarrow 2N, 4C \rightarrow 1N, 2C \rightarrow 1N, 1C$), reductional before equational, an order forced by ordered two-stage REC8 cohesin release (arm separase 0.684 $\rightarrow$ centromeric 1.085, set by the sign of shugoshin protection, not a tuned rate) [V]. *Whether* gametes are identical: no — independent assortment is exactly $2^{23} = 8,388,608$, and crossover interference is the substrate refractory period mapped onto the chromosome axis (sub-Poisson, Fano 0.108 versus the Poisson null 0.976, gap CV 0.26, obligate crossover), so the floor on distinct gametes exceeds 10^{47} [V].

The sperm is a fast oscillator; the egg is a held switch. The flagellar beat is the fast end of the same relaxation-oscillator ladder: in arbitrary units the periods order 26.2 (flagellum) < 132.4 (pulse) < 675.0 (spermatogenic) < 1133.7 (menstrual) [V], anchored to a ~ 20 Hz beat [L]. Hyperactivation is a CatSper-driven rise in oscillator gain, and its two co-signatures move together: the beat grows in amplitude ($2.21 \rightarrow 2.83$) and slows ($77 \rightarrow 57$ beats) [V]. The egg is the opposite use of the kernel: a switch held at metaphase II (arrested state -1.088). A sub-spinodal Ca^{2+} transient does nothing; one above the spinodal (0.3849) flips it irreversibly, and that same past-spinodal irreversibility *is* the polyspermy block [V]. The integer 9 of the 9+2 axoneme is structural and *not* derived; it is declared [O] rather than back-fitted.

A measured γ atlas and an honest null. The gamete-program promoters, in ascending γ : TEKT1 1.3400, NLRP5 1.3475, MOS 1.3619, DAZL 1.3803, CATSPER1 1.4019, DMC1 1.4056, MLH1 1.4085, SPO11 1.4105, PRDM9 1.4165, ZAR1 1.4226, DNAH1 1.4469, ZP3 1.4486, REC8 1.4525. A pre-registered test of whether γ *separates* the three functional modules returns F-like 0.109, $p = 0.617$ — a null, reported as it falls. What is supported, and tested separately, is narrower: the recombination core (SPO11, DMC1, MLH1, PRDM9) is a tight γ cluster (CV 0.0028 versus panel 0.0253, $p = 0.0014$). The framework claims only the second result, at the grade the test supports.

Table 9: Measured gamete-program promoter γ by functional module (NN-stacking ΔG_{37} , SantaLucia 1998). A pre-registered permutation test of module separation returns a null (reported honestly); only the recombination-core cluster is claimed.

module	n	mean γ	sd	range
germline	1	1.3803	0.0000	1.3803–1.3803
meiosis	5	1.4187	0.0173	1.4056–1.4525
sperm	3	1.3963	0.0438	1.3400–1.4469
oocyte	4	1.3951	0.0418	1.3475–1.4486

One substrate, two gametes. The same kernel is a free-running oscillator for the sperm and a held switch for the egg; the count asymmetry has the same root. Spermatogenesis divides symmetrically into 4 equal gametes, while oogenesis divides asymmetrically into 1 large egg and 3 vanishing polar bodies, concentrating the cytoplasm (egg:polar-body mass $49\times$) and giving a measured egg:sperm volume ratio of $\sim 18,963\times$ [L]. Anisogamy — the deepest asymmetry in reproduction — is on this substrate an oscillator and a switch built from one rule; the evolutionary “why” is game-theoretic and is marked [O] (§16).

12 The embryo: from two gametes to a fetus

Answer. The two gametes meet and a fetus is built on the same substrate, each step a discriminant against the vendored R19 switch and FHN oscillator, reusing the gamete results and adding a measured developmental master-gene γ panel as the only new input. *Syngamy*: the sperm (a free oscillator) delivers a supra-spinodal Ca^{2+} kick that flips the egg (a held switch) one-way, with a polyspermy block; two haploid pronuclei fuse, restoring diploidy ($1N+1N \rightarrow 2N$), and the new genome is unique to a floor of 10^{94} — the square of the per-gamete floor, since the zygote is the product of two independent draws [V]. This is the “emerge DNA” step: a new diploid genome that has never existed before.

Cleavage and genome activation. Unlike the asymmetric oocyte division, cleavage is symmetric — 2^n equal blastomeres — and it *subdivides a fixed cytoplasmic mass* rather than growing it (summed mass conserved; each blastomere $1/2^n$ of the zygote) [V]. The first cell-fate decision, inner cell mass versus trophectoderm, is one R19 bistable switch. Zygotic genome activation is the genome’s own switch-on: before it, the embryo runs on maternal oocyte stores (ZAR1, the oocyte-to-embryo factor; NLRP5; c-Mos), with the zygotic genome off; a rising competence drive crosses the spinodal 0.3849 and the genome switches on in one discontinuous step (jump 1.655 at drive 0.3913) [V] — the maternal-to-zygotic handoff from the gamete program this package owns to the embryo’s own genome.

The gene-clock builds the body. With the genome on, the body plan unfolds in the order the cited morphogenesis gene-clock dictates: emergence order = argsort of the spinodal over the master genes, which (the spinodal being monotone in γ) is simply γ ascending. The measured panel orders as $\text{NANOG} \rightarrow \text{SOX17} \rightarrow \text{GATA4} \rightarrow \text{HOXB4} \rightarrow \text{PAX3} \rightarrow \text{CDX2} \rightarrow \text{SOX2} \rightarrow \text{SOX9} \rightarrow \text{PDX1} \rightarrow \text{POU5F1} \rightarrow \text{HOXA1} \rightarrow \text{MYOD1} \rightarrow \text{FOXA2} \rightarrow \text{PAX6} \rightarrow \text{NKX2-5} \rightarrow \text{TBXT} \rightarrow \text{HOXA13}$. A pre-registered test finds γ rises with developmental stage — the stage means climb from pluripotency (1.4310) through germ-layer (1.4373) to organ-primordium (1.4916), Spearman $\rho = 0.551$, permutation $p = 0.020$ [V]. A second pre-registered test ($\text{HOX } 3' \rightarrow 5'$ colinearity) is reported as a *partial* result in the honest style of the gamete chapter: the posterior-most HOX (HOXA13) carries the highest γ , but across three genes the correlation ($\rho = 0.500$, $p = 0.500$) does not reach significance; only the narrow claim is made.

Table 10: Measured developmental master-gene promoter γ (same NN-stacking pipeline; validated on the SOX9 and DAZL anchors), in gene-clock order (argsort of the spinodal = γ ascending). Stage is the pre-registered hypothesis label, declared before γ was seen.

master	stage	γ	func. spinodal	rel. size
NANOG	pluripotency	1.3479	0.6023	1.4226
SOX17	germ layer	1.3821	0.6254	1.4771
GATA4	germ layer	1.3933	0.6330	1.4951
HOXB4	AP-axis (HOX)	1.4453	0.6688	1.5796
PAX3	germ layer	1.4479	0.6706	1.5839
CDX2	germ layer	1.4500	0.6720	1.5873
SOX2	pluripotency	1.4576	0.6773	1.5998
SOX9	organ primordium	1.4598	0.6789	1.6034
PDX1	organ primordium	1.4732	0.6882	1.6255
POU5F1	pluripotency	1.4874	0.6982	1.6491
HOXA1	AP-axis (HOX)	1.4923	0.7017	1.6573
MYOD1	organ primordium	1.4937	0.7027	1.6596
FOXA2	organ primordium	1.4986	0.7061	1.6678
PAX6	organ primordium	1.5110	0.7149	1.6885
NKX2-5	organ primordium	1.5130	0.7163	1.6919
TBXT	germ layer	1.5130	0.7163	1.6919
HOXA13	AP-axis (HOX)	1.5427	0.7375	1.7419

The whole arc. Run end-to-end, a specific sperm and a specific egg produce a zygote with a reproducible, unique genome fingerprint, and the arc passes at every stage (syngamy, cleavage, genome activation, gene-clock body plan). The result is a fetus carrying a unique diploid genome (10^{94}) and an inner-cell-mass-derived body of 17 gene-clock-ordered structures [V]. The absolute gestational timing is left open [O] — the events and their order are forced, but mapping them to a calendar needs external calibration, exactly as puberty’s age does. The full multi-organ morphogenesis atlas and the gene-clock law remain the 4D DNA Blueprint package’s single source of truth, cited here, never re-owned (§16).

13 Infertility versus subfertility: two sides of one spinodal

Answer. The reproductive arc is a chain of substrate operations, and conception requires *every* one to succeed — a logical AND over the 8 verified links of the earlier chapters (the GnRH pulse, the follicle/sperm switch, ordered REC8 meiosis, gamete number, the metaphase-II flip, syngamy, ZGA, the gene-clock), so breaking any single link gives sterility [V]. The two clinical failure classes are then one distinction on the substrate. *Infertility* is an operation on the wrong side of a spinodal: categorical and deterministic, a per-cycle conception probability of exactly zero that a sub-threshold drive cannot move. *Subfertility* is an operation near its threshold: a small but finite Kramers crossing rate per cycle, so the same drive that does nothing across a spinodal moves it exponentially.

The central dissociation. Computed on the same R19 spinodal and Kramers factor $\exp(-\Delta V/D)$ [6] the oncology chapter uses, a fixed, modest drive nudge ($\Delta h = 0.0770$, one fifth of the spinodal 0.3849) raises a near-threshold cycle probability from 0.0391 to 0.0610 ($1.56\times$; over a year $0.381 \rightarrow 0.530$), while leaving a past-spinodal operation at exactly 0 — the same nudge, exponential versus nothing [V]. This is why the two diagnoses behave categorically differently under the same intervention, not as two points on one slider.

Table 11: The infertility/subfertility dissociation under an *identical* drive nudge ($\Delta h = 0.0770$). A near-threshold (subfertile) operation responds exponentially through the Kramers rate [6]; a past-spinodal (sterile) operation stays at exactly zero. Same drive, two regimes.

operation	p_{cycle}	p_{cycle} nudged	fold	p_{year}	p_{year} nudged
subfertile (near threshold)	0.0391	0.0610	$1.56\times$	0.381	0.530
sterile (past spinodal)	0.0000	0.0000	—	0.000	0.000

Male and female factor on the kernel. Male factor is oscillator throughput: whether spermatogenesis runs at all is an R19 switch (sub-spinodal = azoospermia, categorical and sterile) [V], while *how much* it makes once on is the seminiferous/flagellar beat rate — a fast beat (full count) versus a slow one (oligo-/astheno-, subfertile but non-zero: $51 \rightarrow 7$ beats) [V]; CatSper (measured $\gamma = 1.4019$) is a *separate* supra-spinodal gate [3]. Female factor adds the molecular clock of female subfertility: anovulation is the mid-cycle ovulatory-surge switch left OFF (sterile until driven across) [V], and REC8 cohesin fatigue (measured $\gamma = 1.4525$) is the rising-with-age mis-segregation clock — as the held barrier $\gamma^2/4$ decays with the years held, per-egg aneuploidy escape climbs from 0.121 at age 25 to 0.752 at age 45 (only the rising *shape* is claimed; the absolute fraction is [O]), with primary ovarian insufficiency the same latch lost early [V].

Treatment is a drive. The interventions are reading the same kernel: the hCG trigger / ovulation induction is a supra-spinodal kick forcing the stuck ovulatory switch across (the spinodal kick the disease chapter already retrodicts) [V]; pulsatile GnRH *restarts* the oscillator where a continuous drive suppresses (41 versus 0 pulses, cited from the temporal-pattern chapter, same molecule opposite effect) [V]; and ICSI/IVF supply or bypass the missing gate (the same supra-spinodal flip) [V]/[L]. The absolute per-cycle probabilities, the map from a real semen parameter or maternal age to a drive value, and the gene-to- γ links for specific azoospermia/POI genes remain open [O] (§16); sex-hormone levels stay this package’s single source of truth, and the management of a real couple’s infertility belongs to clinicians.

14 Sex determination and sex-ratio distortion: a tiltable coin

Answer. Mammalian sex is one bistable switch: SOX9 (testis, measured $\gamma = 1.4598$) against FOXL2 (ovary, $\gamma = 1.4829$), with SRY the tipping drive. Mendelian 50:50 is that switch untilted — a fair coin. A meiotic-drive gene tilts it, so transmission climbs from one-half and, past the spinodal, fixes near one. A sex-chromosome-linked driver tilts the X-versus-Y gamete switch, skewing offspring sex — male for an X-shredder, female for a Y-killer — while Fisher’s restoring force holds the *population* at 1:1 [10]. Sex is a coin that can be weighted; the genome and the population each set a different constraint.

Sex is a switch; segregation is a fair coin. The SOX9 $\leftrightarrow$ FOXL2 mutual-antagonism toggle reduces to one R19 well of scale $\gamma = 1.4714$ (the mean of the two measured antagonist γ , not fitted), spinodal 0.6869. At zero tilt the two basins have equal depth (both sexes viable), the undifferentiated default settles to ovary, and SRY is a *transient* supra-spinodal drive: it flips the switch to testis and the basin then self-maintains for life (hysteresis) [V]. A heterozygous transmission locus is the same switch with no tilt, so an ensemble of 20,000 meioses transmits at 0.4983 (analytic exactly 0.5) — Mendel’s first law as an untilted R19 switch [V].

Drive is a tilt; the sex ratio is a *signed* tilt. A meiotic-drive gene is a tilt h on the switch, and the transmission ratio is a Boltzmann basin occupancy: it rises monotonically from 0.5 through 0.8623 (graded distortion below the spinodal) to 1.0 once the loser basin vanishes past the spinodal (the t-haplotype / SD fixation limit) [V]. A sex-chromosome-linked gamete-killer

is the same tilt *with a sign*: an X-shredder kills X-bearing sperm and gives a secondary sex ratio of 1.0 (male), a Y-killer gives 0.0 (female), and a weaker driver gives a partial skew (0.8223) — one mechanism, two signs [V].

Fisher holds the line. Drive tilts the individual switch, but at the population level the 1:1 sex ratio is a stable attractor: the rarer sex has higher per-capita reproductive value, so selection favours producers of the rarer sex and suppressors of drive [10]. Displaced to 0.50 or 0.50 male the population returns to 0.5 with no drive, and a persistent weak drive shifts the fixed point only boundedly (to 0.5125, not to fixation) [V]. The human at-birth ratio (~ 0.512 male) is exactly this kind of small residual, *not* a strong-drive case; its cause (paternal age, hormonal-timing) is left open [O].

Table 12: Measured sex-determination master-gene promoter γ (same NN-stacking ΔG_{37} pipeline [3, 2]; validated on the SOX9 and FOXL2 anchors), ascending. A pre-registered permutation test of whether the gonadal axis separates γ returns a null (reported as it falls).

master	axis	γ	GC
SRY	testis	1.2550	0.361
DMRT1	testis	1.4423	0.532
SOX9	testis	1.4598	0.545
RSPO1	ovary	1.4669	0.561
FOXL2	ovary	1.4829	0.575
WNT4	ovary	1.5992	0.672

The sex-determination masters are measured through the identical DNA pipeline (testis axis SRY/SOX9/DMRT1, ovary axis FOXL2/RSPO1/WNT4). A pre-registered test of whether the gonadal axis *separates* promoter γ — declared before the values were seen, exactly as in the gamete chapter — finds the ovary axis trends higher (mean 1.5163 versus testis 1.3857, gap 0.1306), but with SRY a strong low- γ outlier (1.2550, AT-rich) and $n = 3$ per axis the permutation test returns $p = 0.10$ — a null, reported honestly [V]. None of the chapter’s verified claims depends on this test; the measured γ sets the switch *scale*, not the drive strength (§16).

15 Methods: reproducibility, determinism, and grading

Determinism (VP-SPEC C1). The reproduction engine is seeded (SEED = 19) with single-threaded BLAS, round-before-hash, and sorted JSON keys, so the hashed core emits an identical sha256 on repeat: f849da7d13d233020b3f4051867307c7523c42f9ae485359113a33b55f6a94a4 (two runs byte-identical, process-stable). Every quantity in this paper is pulled live from the research modules at build time by the LaTeX generator (`tools/build_paper.py`); nothing is transcribed by hand, so the PDF cannot drift from the engine. The companion HTML site is generated the same way (`tools/build_docs.py`), and two full site builds are byte-identical.

No tuning (strict). Every constant is a measured input or a derived value, never chosen to hit a target. The recovery timescale τ_s sets a period, but periods are [L] anchors — the package never claims to *derive* a clinical timing. The noise constant D , per-year rates, coupling κ , and amplitude fractions are stated inputs; all RR ratios are invariant to D . The master-gene γ values are *measured* from cached promoters (nearest-neighbour ΔG_{37} , SantaLucia 1998) and accepted only because the pipeline reproduces the vendored SOX9 anchor bit-for-bit.

Research gate. All sections pass one machine-checked gate before the writing phase is unlocked: determinism ✓, organ emergence ✓, the T1–T5 discriminant battery ✓, oncology ✓, therapy ✓, and disease ✓ (`all_green = true`).

Grading vocabulary. [V] simulation-verified; [F] forced by geometry; [L] measured anchor; [O] open, with the obstacle stated in §16. The grades are honesty markers, not self-deprecation: the physics fixes shape, ordering, and ratios; absolute scales and external calibrations are declared open rather than tuned.

16 Open quantities (honest grading)

The physics fixes the shape, ordering, and ratios; the following absolute scales and externally-set calibrations are declared open rather than tuned. Genuinely novel clinical hypotheses (low-frequency pulsatile restoration in PCOS; pulsed/cycled testosterone replacement) are likewise [O] and require clinical validation. This is a physics-derived research artefact and is **not** medical advice.

Table 13: Every open [O] quantity with its specific obstacle (VP-SPEC C3). An open quantity without a stated obstacle is a gate failure.

open quantity	grade	obstacle (why not reproducible in-package)
absolute organ size / mass	[O]	DWELL $\propto \gamma^{3/2}$ fixes relative size and order [F]; absolute scale needs external calibration
absolute cancer incidence	[O]	RR <i>shape</i> reproduces [V]; absolute population incidence needs epidemiological calibration
noise level D (abs. RR magnitude)	[O]	single open lattice constant; every reported RR <i>ratio</i> is invariant under D (verified D -robust)
chronological pubertal age	[O]	the spinodal crossing fixes discontinuous order and jump/spinodal ratio [V]; calendar age needs endocrine calibration
menopause irreversibility	[O]	latch collapse on depletion is shown [V]; no in-package follicle regeneration, so one-wayness is asserted from biology
absolute gestational timing	[O]	fertilisation $\rightarrow$ fetus events and their <i>order</i> are forced [V]; mapping to days/weeks needs external calibration (as pubertal age)
full-body morphogenesis atlas	[O]	the gene-clock law (argsort spinodal) and the multi-organ atlas are the 4D DNA Blueprint package's SSOT (firewall); this package demonstrates the cited clock on a panel only
absolute per-cycle conception probability	[O]	the categorical/probabilistic dissociation and the rising age curve are verified [V]; absolute rates for a real couple need clinical calibration
per-age aneuploidy fraction	[O]	only the rising <i>shape</i> of REC8 cohesin-fatigue mis-segregation is claimed [V]; the absolute fraction and the age $\rightarrow$ cohesin map need calibration
meiotic-drive / sex-ratio tilt magnitude	[O]	drive-as-tilt and the signed secondary-sex-ratio skew are verified [V]; the $\gamma \rightarrow$ drive-strength map per real driver is not derived
cause of the human ~ 0.512 sex ratio	[O]	the 1:1 Fisher attractor and the bounded residual are verified [V]; the cause (paternal age / hormonal-timing) is unpinned
axis separation of sex-determination γ	[O]	the atlas is measured [V]; whether the gonadal axis separates γ at $n = 3$ is reported ($p = 0.10$, a null), not assumed

Data and code availability

All code, the deterministic reproduction engine, the cached promoter sequences, and the generators for this paper and the companion HTML site are bundled in the reproducibility archive deposited with this record (DOI [10.5281/zenodo.20754657](https://doi.org/10.5281/zenodo.20754657); Zenodo concept DOI, version-independent and always resolving to the latest version). The living canonical version is maintained at jamming-physics.org/reproductive-endocrine/, with the public mirror at [GitHub](https://github.com). Running `python repro/run_all.py` reproduces every number; running `python tools/build_paper.py` regenerates this document and `python tools/build_docs.py` regenerates the site.

Acknowledgement of scope

This paper derives the gonadal-endocrine axis only; the brain-facing HPA stress axis and felt experience are firewalled to the companion Felt Cognition paper. Nothing here is clinical advice.

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